

Customer Churn Exploratory Data Analysis Report

Submitted By

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Task: Task 3 - Custom Churn - Exploratory Data Analysis

Dataset: IBM Telco Customer Churn Dataset

1. Introduction

Customer churn refers to customers who stop using a company's services. Understanding the factors that influence churn helps businesses improve customer retention, reduce revenue loss, and enhance customer satisfaction.

The objective of this project is to perform Exploratory Data Analysis (EDA) on a customer churn dataset to identify important patterns and business insights.

2. Dataset Details

Dataset Name:

IBM Telco Customer Churn Dataset

Source:

Kaggle

Rows:

7043

Columns:

21

Target Variable:

Churn

The dataset contains customer demographic information, account details, service subscriptions, billing information, and churn status.

3. Dataset Overview

The dataset includes:

- Customer demographics
- Internet services

- Phone services
- Contract type
- Payment method
- Monthly charges
- Total charges
- Customer tenure
- Churn status

The target variable indicates whether a customer has churned or not.

4. Data Cleaning

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked for missing values.
- Converted the **TotalCharges** column to numeric format.
- Filled missing values using the median.
- Checked for duplicate records.
- Removed the **CustomerID** column as it is not useful for analysis.
- Verified data types for all columns.

After cleaning, the dataset was ready for analysis.

5. Exploratory Data Analysis

Several statistical analyses and visualizations were performed.

Statistical Summary

The numerical features were analyzed using:

- Mean
- Median
- Standard Deviation
- Minimum

- Maximum

These statistics helped understand the overall distribution of the dataset.

6. Visualizations

Chart 1 – Customer Churn Count

A count plot was created to compare churned and non-churned customers.

Observation

Most customers remained with the company, while a smaller percentage churned.

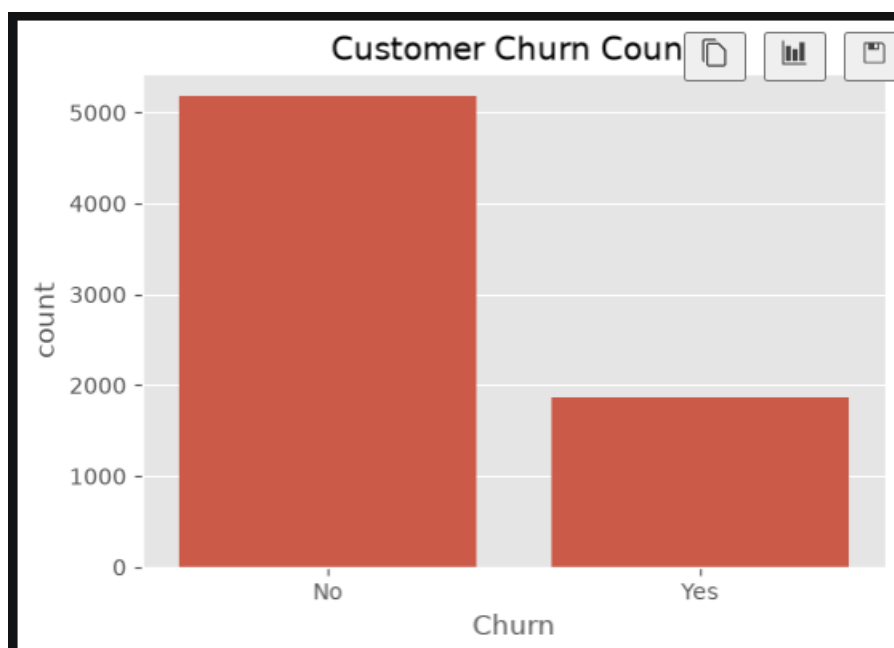


Chart 2 – Churn Percentage

A pie chart was used to visualize the proportion of churned customers.

Observation

Approximately one-fourth of customers left the company.

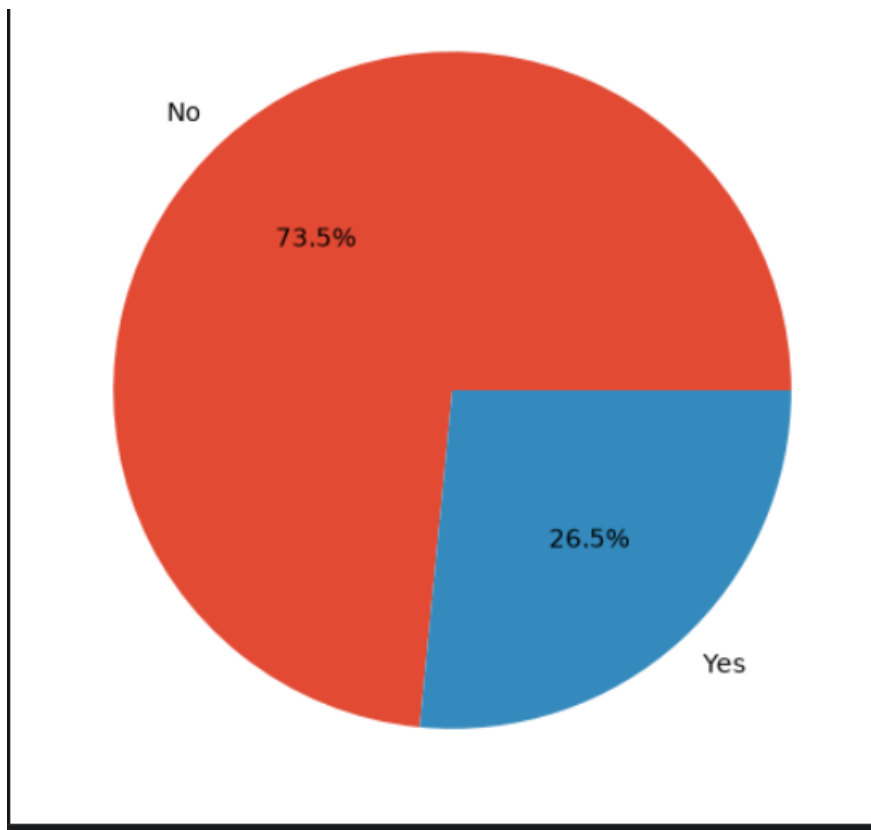


Chart 3 – Tenure Distribution

A histogram was plotted for customer tenure.

Observation

Many customers have relatively short tenure, while fewer customers stay for very long periods.

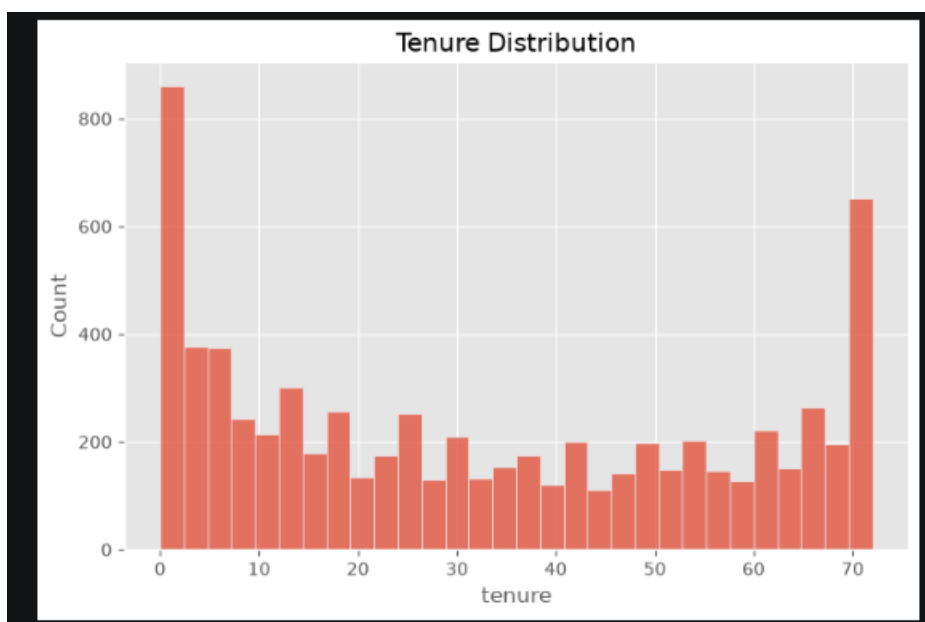


Chart 4 – Monthly Charges Boxplot

A boxplot was used to detect outliers.

Observation

Monthly charges contain several high-value outliers.

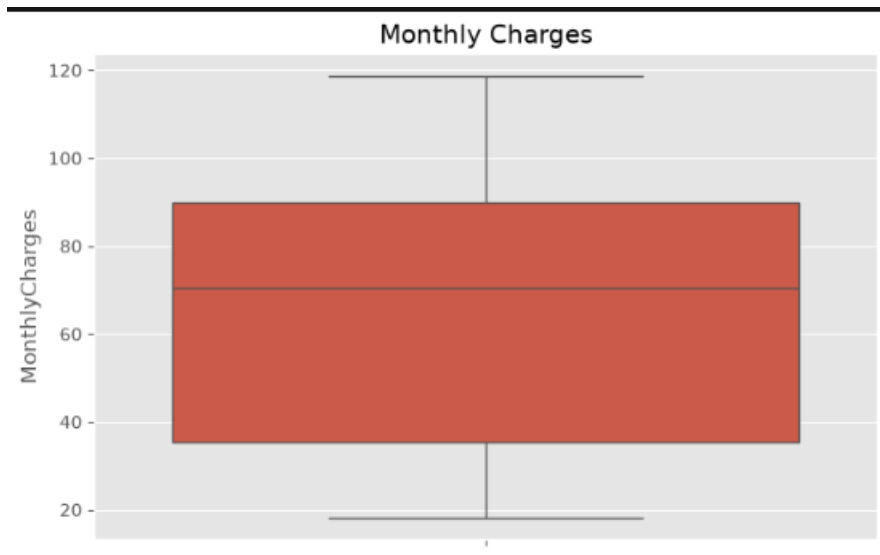


Chart 5 – Contract Type Distribution

A count plot compared different contract types.

Observation

Most customers prefer month-to-month contracts.

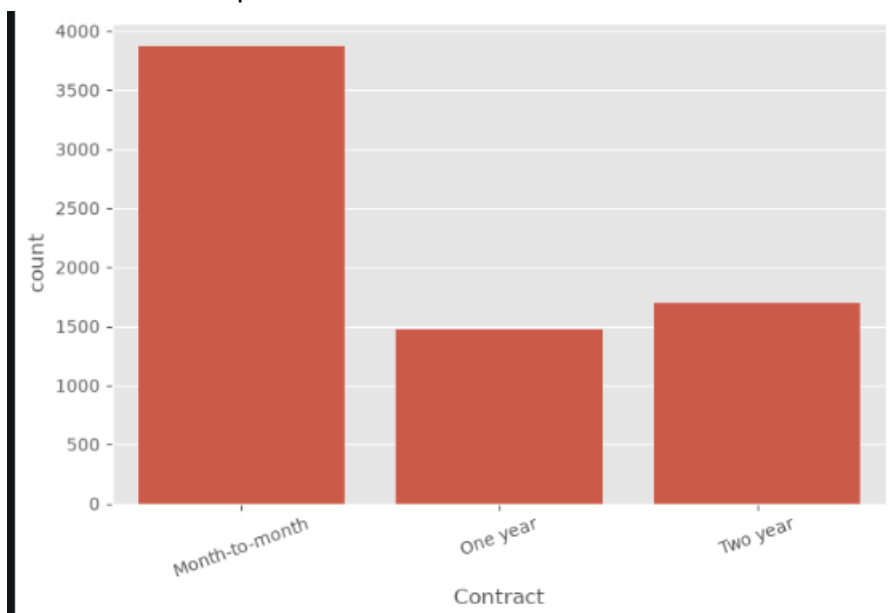


Chart 6 – Contract Type vs Churn

A count plot with churn comparison was created.

Observation

Customers with month-to-month contracts exhibit the highest churn rate.

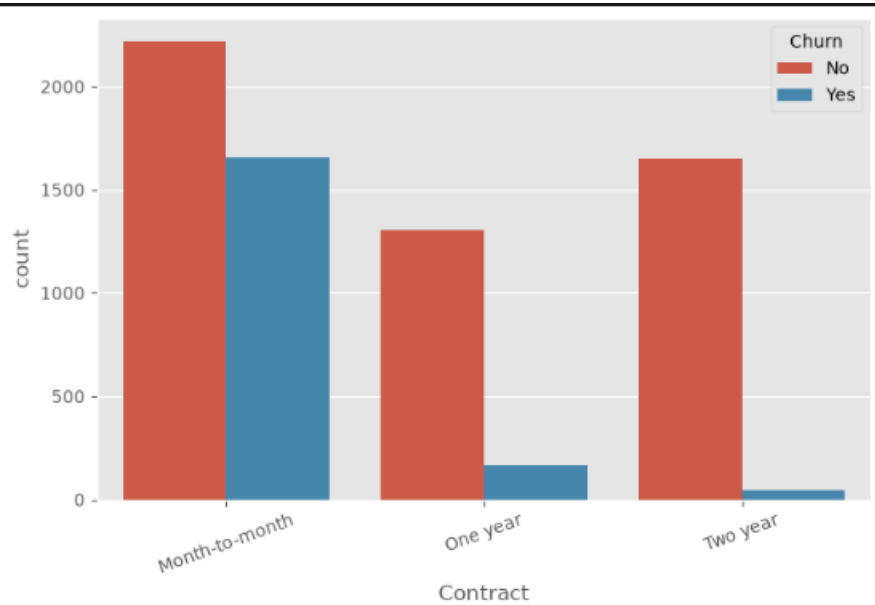


Chart 7 – Correlation Heatmap

A heatmap was generated for numerical features.

Observation

Tenure and Total Charges have a strong positive correlation.

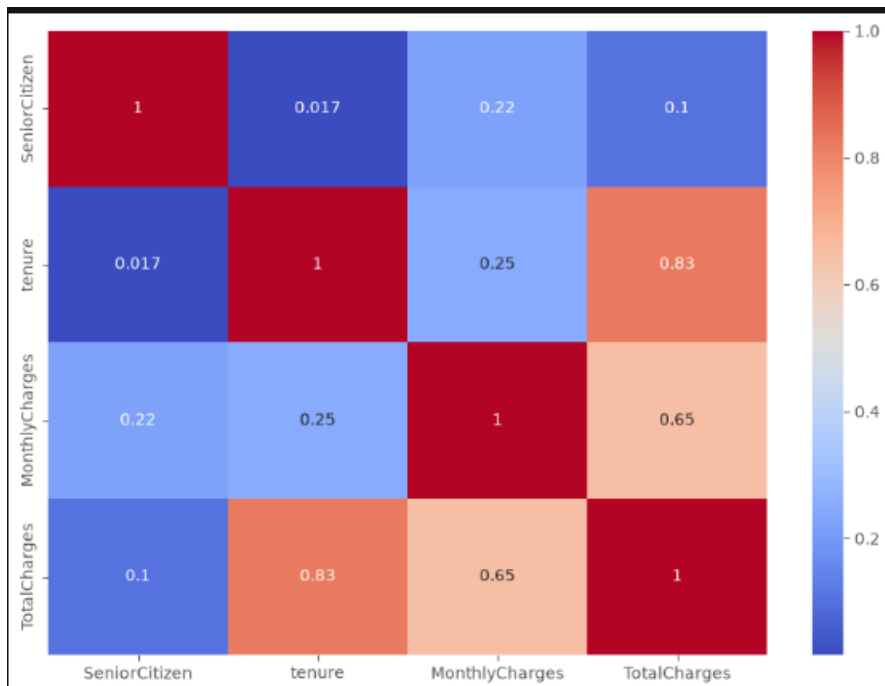
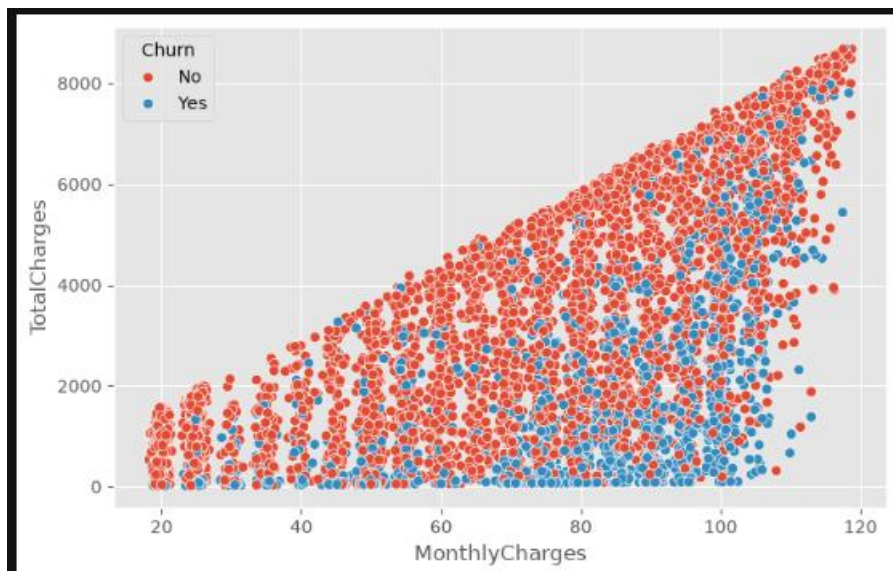


Chart 8 – Monthly Charges vs Total Charges

A scatter plot analyzed the relationship between two numerical variables.

Observation

Customers with longer tenure generally accumulate higher total charges.



7. Key Findings

1. Around **26%** of customers churned.

2. Customers with month-to-month contracts have the highest churn rate.
 3. Customers with shorter tenure are more likely to leave.
 4. Higher monthly charges are associated with increased churn.
 5. Tenure and Total Charges are strongly correlated.
 6. No major missing value issues remained after preprocessing.
 7. Contract type, tenure, and monthly charges are the most influential factors affecting churn.
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8. Business Recommendations

- Encourage customers to choose long-term contracts.
 - Offer loyalty rewards for customers with shorter tenure.
 - Provide discounts to customers with high monthly charges.
 - Improve customer support for high-risk customer groups.
 - Conduct periodic customer satisfaction surveys.
 - Introduce personalized retention offers before contract renewal.
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9. Conclusion

The exploratory data analysis identified several important factors influencing customer churn. Contract type, tenure, and monthly charges play a significant role in customer retention.

The insights generated from this analysis can help organizations reduce churn through targeted retention strategies, improved customer engagement, and better service offerings.
